Project Title

News & Social Media Analytics Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by
Harshitha Bollineni (2520080007)
Naeeni Satwika (2520090151)

Team Number:
Section: 12

Under the guidance of

Chandusha Kanda

Ms. K. Chandusha, Asst. Prof, CSIT

Hyderabad-500090, Telangana, India.

ABSTRACT

The News & Social Media Analytics Engine is a DSA-based system designed to efficiently process and analyze large volumes of news articles and social media posts. The system helps identify duplicate content, search relevant information, cluster related stories, assign moderators, and manage large-scale data processing. It aims to reduce information redundancy and improve the organization and retrieval of digital content.

The project applies advanced Data Structures and Algorithms, including KMP and Rabin-Karp for fast pattern searching, suffix arrays and LCP for document similarity, and Dynamic Programming for fuzzy matching and query correction. These techniques enable accurate and efficient analysis of textual data.

To handle large workloads, the system can incorporate network-flow techniques for assignment and resource allocation, randomized algorithms, and parallel processing for improved scalability and performance. Overall, the project demonstrates the practical application of advanced DSA concepts in building an efficient, scalable, and intelligent platform for news and social media content analysis. Furthermore, this scalable system provides critical real time analytics to optimize data delivery networks worldwide.

Signature of the Guide